Representative Structural Calculation Package

Transfer podium frame

Item	Value
Project	Transfer podium frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	09-transfer-podium-frame

Nodes	Members	Loads	Combos	Max disp.	Max util.
81	**158**	**448**	**28**	**37.28 mm**	**2.652**

Model And Design Basis

Item	Value
Building type	transfer
Description	Six-story frame with a two-story podium and narrower upper block.
Occupancy	Office
Bounds X/Y/Z	28.00 / 14.00 / 21.00
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

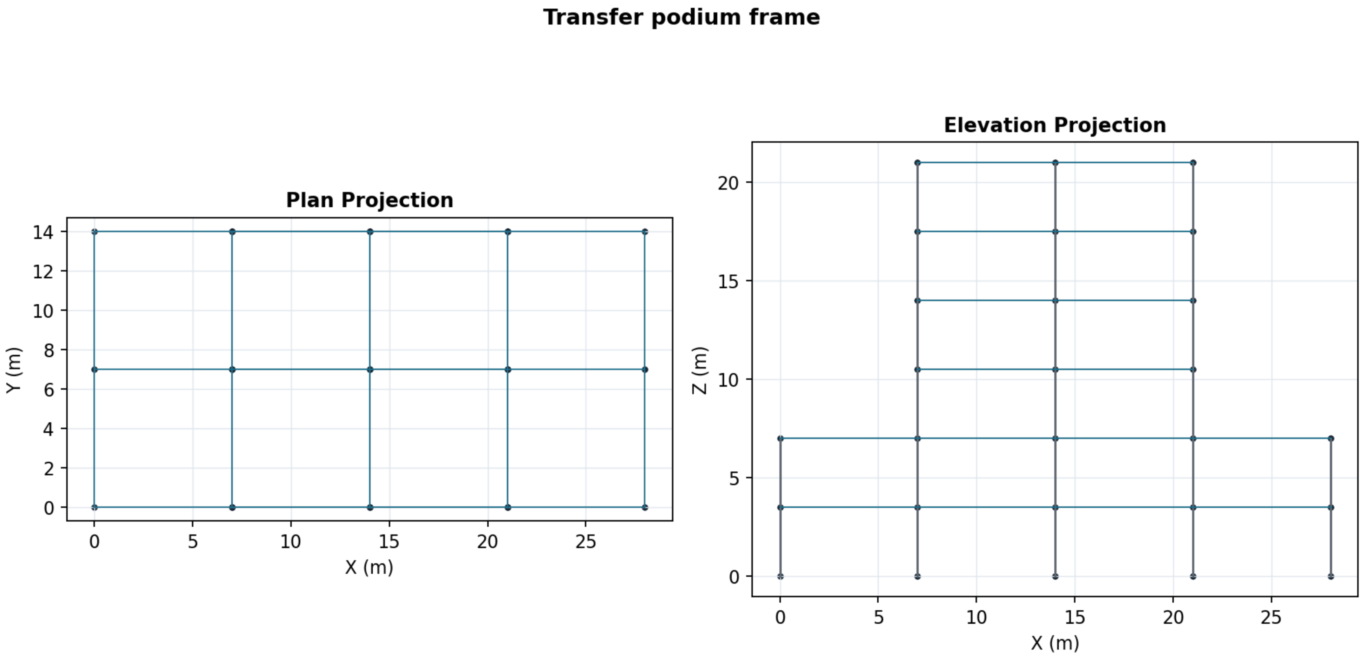
Case	Type	Loads	Fx	Fy	Fz
D	dead	92	0.000 kN	0.000 kN	-7997 kN
L	live	92	0.000 kN	0.000 kN	-3665 kN
WX	wind	66	132.30 kN	0.000 kN	0.000 kN
WY	wind	66	0.000 kN	264.60 kN	0.000 kN
EX	seismic	66	811.09 kN	0.000 kN	0.000 kN
EY	seismic	66	0.000 kN	811.09 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	392.00	1333 kN	666.40 kN	22.05 kN	44.10 kN
2	392.00	1333 kN	666.40 kN	22.05 kN	44.10 kN
3	392.00	1333 kN	666.40 kN	22.05 kN	44.10 kN
4	392.00	1333 kN	666.40 kN	22.05 kN	44.10 kN
5	392.00	1333 kN	666.40 kN	22.05 kN	44.10 kN

6	392.00	1333 kN	333.20 kN	22.05 kN	44.10 kN

Geometry



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	5.280 mm	0.410	0.000
KDS-ST-02	OK	6.779 mm	0.563	0.000
KDS-ST-04-WX-P	OK	7.090 mm	0.529	0.000
KDS-ST-04-WX-N	OK	7.090 mm	0.529	0.000
KDS-ST-04-WY-P	OK	10.97 mm	0.614	0.000
KDS-ST-04-WY-N	OK	10.97 mm	0.614	0.000
KDS-ST-05-EX-P	OK	31.67 mm	0.768	0.000
KDS-ST-05-EX-N	OK	31.67 mm	0.768	0.000
KDS-ST-05-EY-P	OK	37.28 mm	0.921	0.000
KDS-ST-05-EY-N	OK	37.28 mm	0.921	0.000
KDS-ST-06-WX-P	OK	5.153 mm	0.309	0.000
KDS-ST-06-WX-N	OK	5.153 mm	0.309	0.000
KDS-ST-06-WY-P	OK	9.834 mm	0.395	0.000
KDS-ST-06-WY-N	OK	9.834 mm	0.395	0.000
KDS-ST-07-EX-P	OK	31.30 mm	0.560	0.000
KDS-ST-07-EX-N	OK	31.30 mm	0.560	0.000
KDS-ST-07-EY-P	OK	36.97 mm	0.701	0.000

KDS-ST-07-EY-N	OK	36.97 mm	0.701	0.000

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M62	NG	2.652	steel-deflection	KDS-ST-05-EX-N
2	M59	NG	2.643	steel-deflection	KDS-ST-05-EX-P
3	M65	NG	2.643	steel-deflection	KDS-ST-05-EX-P
4	M63	NG	2.638	steel-deflection	KDS-ST-05-EY-N
5	M61	NG	2.638	steel-deflection	KDS-ST-05-EY-N
6	M60	NG	2.633	steel-deflection	KDS-ST-05-EY-N
7	M66	NG	2.633	steel-deflection	KDS-ST-05-EY-P
8	M58	NG	2.633	steel-deflection	KDS-ST-05-EY-N
9	M64	NG	2.633	steel-deflection	KDS-ST-05-EY-P
10	M53	NG	2.420	steel-deflection	KDS-ST-05-EX-P
11	M56	NG	2.415	steel-deflection	KDS-ST-05-EX-P
12	M50	NG	2.415	steel-deflection	KDS-ST-05-EX-N
13	M54	NG	2.401	steel-deflection	KDS-ST-05-EY-P
14	M52	NG	2.401	steel-deflection	KDS-ST-05-EY-N
15	M57	NG	2.397	steel-deflection	KDS-ST-05-EY-P
16	M51	NG	2.397	steel-deflection	KDS-ST-05-EY-N
17	M49	NG	2.397	steel-deflection	KDS-ST-05-EY-N
18	M55	NG	2.397	steel-deflection	KDS-ST-05-EY-P
19	M44	NG	2.033	steel-deflection	KDS-ST-05-EX-N
20	M47	NG	2.027	steel-deflection	KDS-ST-05-EX-N

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	158	102	2.652
Connections	158	62	8.738
Foundations	15	62	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.

- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.